Formal Methods

Lecture # 5

Predicate Logic

(First Order Logic - FOL)

Predicate Calculus

In this chapter we introduce another part of our logical language. The language of propositions introduced in the previous chapter allows us to make statements about specific objects

We require a language that reveals the internal structure of our propositional statements, a language that allows us to take them apart and apply them to objects without proper names. The language we require is the language of predicate calculus.

Syntax of FOL: Basic elements

- Constants KingJohn, 2,...
- Functions Sqrt, Likes...
- Variables x, y, a, b,...
- Connectives $\neg, \Rightarrow, \wedge, \vee, \Leftrightarrow$
- Equality =
- Quantifiers $\forall, \exists$

Predicates

- A predicate is a proposition whose truth depends on the value of one or more variables.
- For example, "n is a perfect square" is a predicate whose truth depends on the value of n.
- A function like notation is used to denote a predicate supplied with specific variable values. P(n)="n is a perfect square"
- P(4) is true and P(5) is false.

Universal Quantification

- Universal Quantification denoted as $\forall$
- Universal Quantification allows us to capture statements of the form “for all” or “for every”.
- For example, “every natural number is greater than or equal to zero” can be written formally as

$$\forall n : N \bullet n \geq 0$$

Universal Quantification

- A quantified statement consists of three parts: the quantifier, the quantification, and the predicate
- Every predicate logic statement can be considered as follows

$$\text{quantifier quantification} \bullet \text{predicate}$$

Universal Quantification

$$\forall p : People; d : Dog \bullet \text{p is owner} \wedge \text{p owns d}$$

Exercise

- Everybody likes Jaffa cakes
- All vegetarians don’t like Jaffa cakes
- Everybody either likes Jaffa cakes or is a vegetarian
- Either every body likes Jaffa cakes or everybody is a vegetarian

Solution

$\forall x : \text{People} \bullet x \text{ likes Jaffa cakes}$

$\forall x : \text{People} \bullet (x \text{ is vegetarian} \wedge \neg(x \text{ likes Jaffa cakes}))$

$\forall x : \text{People} \bullet (x \text{ likes Jaffa cakes} \vee x \text{ is vegetarian})$

$(\forall x : \text{People} \bullet x \text{ likes Jaffa cakes}) \vee (\forall x : \text{People} \bullet x \text{ is vegetarian})$

Universal Quantification

- Law 1

$$((\forall x : X \bullet p) \vee (\forall x : X \bullet q)) \Rightarrow (\forall x : X \bullet p \vee q)$$

- Example

$$((\forall \mathrm{p} : \mathrm{Person} \bullet \mathrm{p\ is\ wise}) \vee (\forall \mathrm{p} : \mathrm{Person} \bullet \mathrm{p\ is\ strong})) \Rightarrow$$
$$(\forall \mathrm{p} : \mathrm{Person} \bullet \mathrm{p\ is\ wise} \vee \mathrm{p\ is\ strong})$$

Universal Quantification

- Law 2

$$((\forall x : X \bullet p) \wedge (\forall x : X \bullet q)) \Rightarrow (\forall x : X \bullet p \wedge q)$$

- Example

$$((\forall \mathrm{p} : \mathrm{Person} \bullet \mathrm{p\ is\ wise}) \wedge (\forall \mathrm{p} : \mathrm{Person} \bullet \mathrm{p\ is\ strong}))$$

$$(\forall \mathrm{p} : \mathrm{Person} \bullet \mathrm{p\ is\ wise} \wedge \mathrm{p\ is\ strong})$$

Existential quantification

- Existential quantifier denoted by $\exists$
- Existential quantification is used to assert that a property holds of some (or at least one) elements of a set
- "Some natural numbers are divisible by 3" may be written as

$$\exists n : N \bullet n \bmod 3 = 0$$

Exercise

- Some people like Jaffa cakes
- Some vegetarians don’t like Jaffa cakes
- Some people either like Jaffa cakes or are vegetarian
- Either some people like Jaffa cakes or some people are vegetarian

Solution

$\exists x : \text{People} \bullet x \text{ likes Jaffa cakes}$

$\exists x : \text{People} \bullet x \text{ is vegetarian} \wedge \neg(x \text{ likes Jaffa cakes})$

$\exists x : \text{People} \bullet (x \text{ likes Jaffa cakes} \vee x \text{ is vegetarian})$

$(\exists x : \text{People} \bullet x \text{ likes Jaffa cakes}) \vee (\exists x : \text{People} \bullet x \text{ is vegetarian})$

Existential quantification

- Law 3

$$(\exists x : X \bullet p \vee q) -> ((\exists x : X \bullet p) \vee (\exists x : X \bullet q))$$

- Example

$$(\exists c : Car \bullet fast(c) \vee small(c)) ->$$
$$((\exists c : Car \bullet fast(c)) \vee (\exists c : Car \bullet small(c)))$$

Existential quantification

- Law 4

$$(\exists x : X \bullet p \land q) \Rightarrow ((\exists x : X \bullet p) \land (\exists x : X \bullet q))$$

- Example

$$\exists c : Car \bullet fast(c) \land small(c)) \Rightarrow$$

$$((\exists c : Car \bullet fast(c)) \land (\exists c : Car \bullet small(c)))$$

Satisfaction and validity

- The predicate n>3 can be considered neither true nor false unless we know the value associated with n

- A predicate p is valid if and only if it is true for all possible values of the appropriate type. That is, if a predicate p is associated with a variable x of type X, then p is valid if, and only if,

- Example

$$\forall x : X \bullet p \text{ is true}$$

The predicate $n \geq 0$ is valid as $\forall n : N \bullet n \geq 0$ is equivalent to true

Satisfaction and validity

- A predicate p is satisfiable if and only if it is true for some values of the appropriate type. That is, if a predicate p is associated with a variable x of type X, then p is satisfiable if, and only if ,

$$\exists x : X \bullet p \text{ is true}$$

- Example

The predicate $n > 0$ is satisfiable as $\exists n : N \bullet n > 0$ is equivalent to true

Satisfaction and validity

- A predicate p is unsatisfiable if, and only if, it is false for all possible values of the appropriate type.

- If a predicate p is associated with a variable x of type X, then p is unsatisfiable if, and only if,

$$\forall x : X \bullet p \text{ is false}$$

Satisfaction and validity

- Valid predicates and tautologies are always true
- Satisfiable predicates and contingencies are sometimes true and sometimes false
- Unsatisfiable predicates and contradictions are never true

The negation of quantifiers

- The statement “some body like Brian” may be expressed via predicate logic as

$$\exists p : Person \bullet p \text{ likes Brian}$$

- To negate this expression, we may write as,

$$\neg\exists p : Person \bullet p \text{ likes Brian}$$

- which in natural language may be expressed as “nobody likes Brian”

The negation of quantifiers

- Logically saying “nobody likes Brian” is equivalent to saying “everybody does not like Brian”.
- The negation of quantifiers behaves exactly in this fashion, just as in natural language, “nobody likes Brian” and “everybody does not like Brian” are equivalent so in predicate logic
- And $\neg(\exists p : Person \bullet p \text{ likes Brian})$

$$\forall p : Person \bullet \neg(p \text{ likes Brian})$$

- are equivalent.

The negation of quantifiers

- Law 3

$$\neg(\exists x : X \bullet p) \Leftrightarrow \forall x : X \bullet \neg p$$
$$\neg(\forall x : X \bullet p) \Leftrightarrow \exists x : X \bullet \neg p$$

- When negation is applied to a quantified expression it flips quantifiers as it moves inwards(i.e negation turns all universal quantifiers to existential quantifiers and vice versa, and negates all predicates)